

Deep Learning Approaches for Automated Caries Detection from Radiographic and Intraoral Images

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Abstract—Deep learning has become a leading technique in medical image analysis, enabling automated and consistent diagnosis from visual data. Within dentistry, it is increasingly applied to detect dental caries in both X-ray radiographs and standard intraoral photographs. This paper reviews current research in the field, emphasizing datasets, neural network architectures, evaluation practices, and the key challenges still limiting clinical deployment. We summarize progress in convolutional and transformer-based approaches and identify gaps, such as limited multimodal datasets and explainability issues. Our planned work aims to develop a unified system capable of analyzing both radiographic and photographic inputs, improving accessibility and interpretability of AI-driven dental diagnostics.

Index Terms—Dental caries detection, Deep learning, Medical imaging, CNN, Vision Transformer, Explainable AI

I. INTRODUCTION

Dental caries remain one of the most common oral health problems, affecting billions of people worldwide. Early diagnosis can prevent complications and expensive treatments, but it usually depends on how carefully a dentist can examine radiographs or surface photos of teeth. This process can vary greatly between practitioners and is often time-consuming. Artificial intelligence (AI) and especially deep learning (DL) promise to reduce this variability by automatically analyzing medical images and helping professionals make faster and more objective decisions.

In the last few years, researchers have explored different ways to use DL for identifying caries from both radiographic and normal dental images. Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) are now the main families of models used in this area. CNNs are good at identifying local image features such as small dark spots or irregular textures that might indicate a cavity. ViTs, on the other hand, are better at understanding the entire image context. Even though results are promising, there are still gaps such as limited data, weak generalization to new clinics, and models that behave like “black boxes.” In our work, we aim to study both radiographic and intraoral data and to design a system that is more adaptable and explainable for real-world use.

II. RELATED WORK AND CURRENT METHODS

Over the last decade, several studies have addressed the use of deep learning for dental diagnostics. Rahman et al. [3]

developed the *Dental OPG X-ray Dataset* containing over 800 panoramic images labeled by experts, covering categories such as healthy teeth, caries, and fractures. Bayrakdar et al. [2] trained CNN-based models on bitewing radiographs, obtaining strong detection accuracy. Lee et al. [1] introduced a segmentation approach based on U-Net that could highlight the exact affected areas, while Mărginean et al. [11] used an ensemble of several U-Nets to improve consistency across samples. Park et al. [17] applied Mask R-CNN to detect enamel defects, which shows that AI can extend beyond simple cavity detection.

Other researchers have tried to understand which architectures work best for this task. Dashti et al. [5] compared CNNs with Transformer-based models and found that Transformers performed slightly better, especially when the training data were small. Object detection models such as Faster R-CNN [8] and YOLOv5 [9] have also been explored because they can point directly to the problem area using bounding boxes rather than only classifying the image as “cariou” or “healthy.”

Transfer learning is another frequent strategy in this field. Since medical image datasets are often small, many studies start from models trained on large-scale natural image datasets such as ImageNet. These pretrained models are then fine-tuned on dental images, helping them adapt faster and avoid overfitting. Data augmentation is almost always used—rotations, brightness adjustments, and noise simulation make the model more robust. Some recent works also use synthetic data generated by GANs, which can artificially expand datasets. However, this is still an emerging idea because generating realistic dental structures is a complex task.

Recently, more researchers have become interested in using intraoral RGB photos, often taken with consumer-grade cameras or smartphones. These images can show the surface of the teeth directly and are easier to obtain than radiographs, though they come with lighting and angle variations. Only a few studies combine both modalities, even though such a combination could reflect real clinical conditions more accurately. This observation motivated our decision to explore a multimodal approach that works with both image types.

III. FINDINGS AND OBSERVATIONS

Across existing literature, we noticed several patterns. CNN-based approaches dominate the field because they are relatively

simple and effective. However, they often struggle with small or unbalanced datasets. Hybrid architectures that mix CNN and Transformer layers are becoming more popular because they can capture both local details and broader contextual cues.

Another important trend concerns explainability. Many AI models can achieve high accuracy, but dentists hesitate to use them if they cannot understand how the decision was made. Visualization tools such as Grad-CAM maps have become standard, allowing users to see which regions influenced the model's output. This builds trust and provides insight into model behavior. Still, these visualizations are mostly qualitative, and future work needs more structured ways to measure interpretability.

We also noticed that evaluation standards vary widely. Accuracy and F1-score are common metrics, but different papers use different test sets and preprocessing steps. This makes comparisons difficult. The community would benefit from common benchmarks and public test datasets, which would make results easier to reproduce and evaluate fairly.

IV. CHALLENGES AND LIMITATIONS

Despite clear progress, there are still several open challenges. The first and most critical is the lack of diverse datasets. Most dental datasets are relatively small and collected in limited geographic areas. This can cause models to perform poorly on patients from other regions or with different dental equipment. Data imbalance is another recurring issue because healthy samples outnumber decayed ones, so the model tends to predict "healthy" more often.

Another limitation is the generalization problem. As shown by Holtkamp et al. [6], models trained in one clinical environment can lose performance when tested elsewhere due to variations in X-ray exposure, lighting, or camera quality. Explainability also remains a bottleneck. Although methods such as Grad-CAM and SHAP [17] provide visual clues, they do not explain the decision-making process completely. For AI systems to be trusted, both dentists and patients must clearly understand why a certain region was flagged as problematic.

Furthermore, most research focuses only on grayscale X-rays. In reality, dentists use multiple cues such as color, texture, and visual context to assess caries. Incorporating RGB photos or even 3D data like cone-beam CT scans could make AI systems more realistic. Combining different types of inputs, therefore, is not only interesting but necessary for a clinically useful tool.

V. OUR PLANNED CONTRIBUTION

Our project builds on these insights and aims to develop a model that detects dental caries from both radiographic (X-ray) and intraoral (color) images. This dual-modality system could make detection more reliable and flexible, especially in situations where one type of image is unavailable. We will start by training standard CNN architectures such as ResNet and VGG16 as baselines and then experiment with Vision Transformers (ViT) to evaluate their ability to generalize better.

For training, we plan to use open datasets such as the Dental OPG X-ray dataset [3], complemented by publicly available intraoral photos. We will apply image augmentation and normalization to reduce overfitting and test several evaluation metrics, including accuracy, recall, and F1-score. Grad-CAM will be used to visualize model attention and verify if predictions correspond to actual lesion areas. Our long-term goal is to build a small but interpretable prototype that can eventually support dentists in diagnosis and education.

If resources allow, we will design a dual-branch neural network where one branch handles X-rays and another handles color photos. The two feature maps will be fused before classification, allowing the model to learn from both modalities simultaneously. We believe this combination could reveal complementary information because X-rays show internal structure, while photos show surface details. Together, they could make predictions both stronger and more transparent.

VI. RESEARCH QUESTIONS

The study is guided by the following research questions:

- **RQ1:** Does combining radiographic and intraoral imagery improve diagnostic accuracy and model generalization for caries detection?
- **RQ2:** How can explainability methods enhance clinicians' understanding and trust in AI-based diagnostic outputs?
- **RQ3:** Can data augmentation and balanced training improve model performance on small and imbalanced dental datasets?

VII. CONCLUSION

To sum up, deep learning has already transformed the way researchers approach dental caries detection. CNN, U-Net, and Transformer models have all proven capable of identifying carious lesions, often with near-human accuracy. However, limited datasets, lack of standard evaluation, and insufficient explainability still limit clinical integration. Our project proposes to bridge some of these gaps by combining two complementary image types, radiographic and intraoral, and focusing on transparency through explainable AI tools.

We expect that this multimodal and interpretable approach will not only improve detection accuracy but also build confidence among dental practitioners. Ultimately, such systems could be integrated into everyday dental software or mobile applications, making preventive screening easier and more consistent. Future collaborations between engineers and dentists will be essential to expand datasets, create shared benchmarks, and move closer to truly reliable AI support in oral healthcare. In the long term, this line of research could also support preventive dentistry, helping patients monitor their oral health more easily through AI-assisted tools and smartphone-based applications.

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